

Papa Kwesi-Kwah Aggrey

Hamilton, US
Mobile: +1 (551) 338-2197
Email: apapakwesikwah@gmail.com
LinkedIn: www.linkedin.com/in/papa-aggrey
Nationality: Ghanaian

Personal Profile

I am a Ph.D. student in Systems Engineering focused on power systems optimization and smart grids. My research explores both centralized and distributed methods for solving optimal power flow problems in large-scale energy networks. I am interested in mathematical modeling, AC optimal power flow, current-injection formulations, multi-area coordination, and optimization algorithms that improve the reliability, scalability, and intelligence of modern power systems.

Education

2025 – present: PhD in Systems Engineering

Stevens Institute of Technology, Hoboken, NJ, 07030

Graduate Research Assistantship

Supervised by Dr Amro M. Farid

GPA: 3.814/4.00

Relevant Coursework: Smart Grid, Decision Making and Risk Analysis, Elements of Operations Research, Probability and Stochastic Processes.

Recent Work: From Matpower to Hetero-functional Graph Theory: Implementing Electric Power System DC Optimal Power Flow

2019 – 2023: B.Sc in Electrical and Electronic Engineering

Kwame Nkrumah University of Science and Technology (KNUST), Kumasi, Ghana

GPA: 3.88/4.00

Honors: Dean's List, KNUST Vice Chancellor's Students' Excellence Award, First Class Honors.

Relevant Coursework: Control Systems, Power Systems Analysis, Power Generation and Supply, Algebra, Numerical Analysis, Linear Electronics, Digital Signal Processing.

Thesis: Forecasting of Hybrid Wind-Solar Energy using Machine Learning.

Key Research Interests: Power Systems Optimization, Transactive Energy.

Experience

Tuesday, June 2, 2026: CSIRO National Energy Analysis Center: Seminar Presentation

Stevens Institute of Technology

- Presented the implementation of DC Optimal Power Flow (DC-OPF) using the Hetero-Functional Graph Theory (HFGT) framework, extending the application of the HFGT Toolbox within the CSIRO research team.
- Demonstrated the translation of conventional MATPOWER-based power-system models into an HFGT-based representation for power-system analysis and optimization.

November 2023 – October 2024: National Service: Teaching/Research Assistant

Kwame Nkrumah University of Science and Technology

- Assisted in teaching undergraduate Electrical Engineering courses to classes of about a 100 students.
- Developed instructional materials and practice problems to reinforce classroom learning.
- Conducted research on Multi-Objective Grey Wolf Optimization using parameter tuning. MATLAB was used to run the model for reducing cost of generation and transmission losses, and maximizing profit.

October 2022 – November 2022: Electrical Engineering Intern

Remotec Limited, Kumasi, Ghana

- Soldered electronic components, including diodes and GSM modules, onto printed circuit boards (PCBs) to assemble six K-Switch units used to replace photocell-based control systems in street lighting applications.
- Conducted a site assessment of the KNUST Recycling Centre to evaluate equipment condition, identify maintenance and service requirements, and prepare a technical report for departmental review.

Academic Publications

Distributed Profit Maximizing Security Constrained IV-AC Optimal Power Flow using ALADIN. Under review. [IEEE Transactions on Smart Grid]
Papa Kwesi-Kwah Aggrey, Amro M. Farid.

Projects/Research

Optimality Gap and Economic Efficiency Analysis of Classical ACOPF and Current-Voltage ACOPF Formulations

LIINES lab, Stevens Institute of Technology

- Built MATLAB/CasADi implementations of classical AC Optimal Power Flow (ACOPF) and Current-Voltage ACOPF (IV-ACOPF) formulations for profit maximizing power system optimization.
- Developed and applied economic comparison metrics to evaluate ACOPF and IV-ACOPF formulations using generation cost, revenue-to-cost ratio, profit margin, objective gap, opportunity loss, and cost-to-serve improvement.
- Analysed how current-voltage OPF formulations can improve numerical solution quality and reduce generation cost relative to locally solved classical ACOPF models.

Skills

Python, MATLAB, Julia
Languages: English

References

Amro M. Farid

Principal Investigator
Department of Systems Engineering
Stevens Institute of Technology
Email: amro.farid@stevens.edu